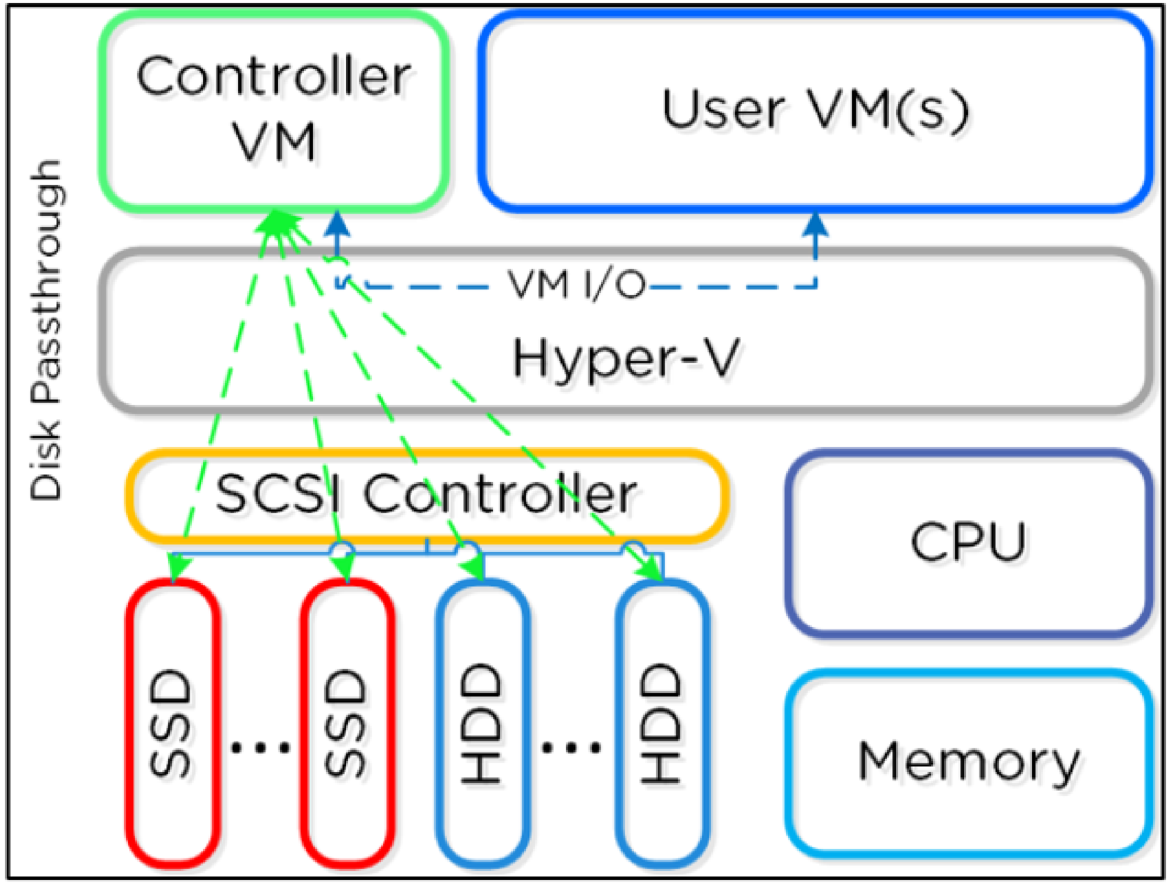
Hyper-V architecture

Overview of Hyper-V

Hyper-V node architecture

When a Nutanix Hyper-V cluster is created, it will automatically join the Hyper-V hosts to the specified Windows Active Directory domain. These hosts are then put into a failover cluster for VM HA. When this is complete, there will be AD objects for each individual Hyper-V host and the failover cluster.

In Hyper-V deployments, the CVM runs as a VM and disks are presented using disk passthrough.



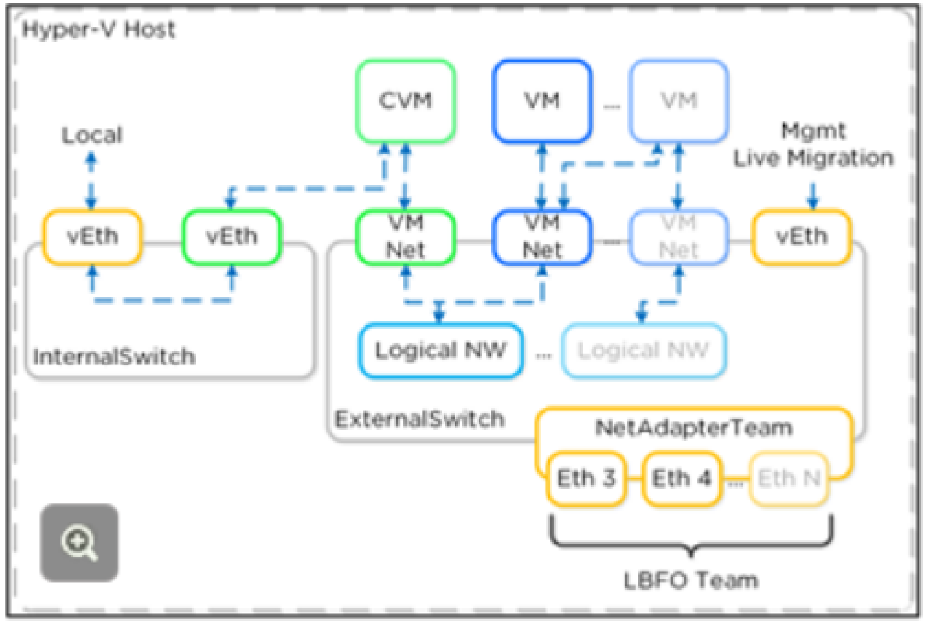
Hyper-V networking architecture

Each Hyper-V host has an internal-only virtual switch, which is used for intra-host communication between the Nutanix CVM and host. For external communication and VMs, an external virtual switch (default) or logical switch is leveraged. The internal switch is for local communication between the Nutanix CVM and Hyper-V host. The host has a virtual Ethernet interface (vEth) on this internal switch (192.168.5.1), and the CVM has a vEth on this internal switch (192.168.5.2). This is the primary storage communication path.

The external vSwitch can be a standard virtual switch or a logical switch. It will host the external interfaces for the Hyper-V host and CVM as well as the logical and VM networks leveraged by VMs on the host. The external vEth interface is leveraged for host management, live migration, and so on. The external CVM interface is used for communication to other Nutanix CVMs. Logical and VM networks can be created as required, assuming the VLANs are enabled on the trunk.

The diagram shows a conceptual overview of the virtual switch architecture.

Click to enlarge the image.



How Hyper-V works on Nutanix

The Nutanix platform supports Microsoft Offloaded Data Transfers (ODX), which allows the hypervisor to offload certain tasks to the array. This is much more efficient because the hypervisor doesn't need to be the "man in the middle." Nutanix currently supports the ODX primitives for SMB, which include full copy and zeroing operations. However, while VAAI has a "fast file" clone operation (using writable snapshots), the ODX primitives do not have an equivalent and perform a full copy. It is therefore more efficient to rely on the native DSF clones which can currently be invoked via nCLI, REST, or PowerShell CMDlets. Currently, ODX is invoked for the following operations:

- In VM or VM to VM file copy on DSF SMB share
- SMB share file copy

How Hyper-V works on Nutanix

Deploy the template from the SCVMM Library (DSF SMB share) – NOTE: Shares must be added to the SCVMM cluster using short names – for example, not FQDN. An easy way to force this is to add an entry to the hosts file for the cluster – for example, `10.10.10.10 nutanix-130`.

ODX is NOT invoked for the following operations:

- Clone VM through SCVMM
- Deploy template from SCVMM Library (non-DSF SMB Share)
- XenDesktop Clone Deployment

Users can check whether ODX operations are taking place by going to the "NFS Adapter" Activity Traces page. The operations activity will show as **NfsSlaveVaaiCopyDataOp** when copying a vDisk and **NfsSlaveVaaiWriteZerosOp** when zeroing out a disk.

Useful Hyper-V administration links

- Hyper-V administration for Acropolis: https://portal.nutanix.com/#/page/docs/details?targetId=HyperV-Admin-AOS-v511:HyperV-Admin-AOS-v511
- Collecting logs for Hyper-V 2016 or 2019: https://docs.microsoft.com/en-us/virtualization/community/team-blog/2018/20180123-looking-at-the-hyper-v-event-log-january-2018-edition
- Collecting logs for Hyper-V use PowerShell https://github.com/MicrosoftDocs/Virtualization-Documentation/tree/live/hyperv-tools/HyperVLogs
- Hyper-V - NIC Troubleshooting https://portal.nutanix.com/#/page/kbs/details?targetId=kA0600000008fTSCAY
- Troubleshooting SMB Access https://portal.nutanix.com/#/page/kbs/details?targetId=kA0600000008i7gCAA
- Hyper-V: Accessing Nutanix Storage https://portal.nutanix.com/#/page/kbs/details?targetId=kA0600000008ZMPCA2
- Hyper-V: Playbook and Getting Started Guide https://portal.nutanix.com/#/page/kbs/details?targetId=kA060000000TSYyCAO